

## SHEBOYGAN, WI, USA

WMO: 994330

Lat: 43.7495N

Lon: 87.6927W

Elev: 176

StdP: 99.23

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -19.2      | -16.5 | -25.2                         | 0.4 | -18.7 | -22.4 | 0.5 | -15.9 | 21.1                  | -0.8 | 17.6 | -1.2 | 7.4                   | 280  | 0.720 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 5.2                    | 28.4            | 21.7 | 26.4 | 21.3 | 24.8 | 20.9 | 24.5                | 26.0 | 23.2 | 24.8 | 22.1 | 23.9 | 6.8                  | 240  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|----------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)            |
| 24.0                            | 19.3 | 25.1 | 22.8 | 17.9 | 24.1 | 21.6 | 16.6 | 23.0 | 74.9          | 26.2 | 70.0 | 24.8 | 65.8 | 24.0 | 29.5           |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 18.6          | 15.3 | 12.8 | DB                         | -22.4 | 32.1               | 3.7 | 2.2                                                | -25.0 | 33.7       | -27.2 | 34.9       | -29.2 | 36.1       | -31.9 | 37.7 | (4) |
| (5)               |      |      | WB                         | -22.1 | 25.8               | 3.1 | 1.8                                                | -24.4 | 27.1       | -26.2 | 28.1       | -28.0 | 29.1       | -30.3 | 30.4 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 7.8         | -5.6  | -4.2  | 0.6   | 5.9   | 10.6  | 16.3  | 20.2  | 20.2  | 16.6  | 10.4  | 4.1   | -2.2  | (6)  |  |
|                                                                        | DBStd       | 10.01       | 6.30  | 5.50  | 4.86  | 3.58  | 3.77  | 3.78  | 3.15  | 2.78  | 3.55  | 3.92  | 4.94  | 5.65  | (7)  |  |
|                                                                        | HDD10.0     | 1953        | 484   | 398   | 292   | 132   | 38    | 1     | 0     | 0     | 2     | 43    | 183   | 379   | (8)  |  |
|                                                                        | HDD18.3     | 4033        | 743   | 631   | 549   | 374   | 241   | 83    | 16    | 12    | 73    | 247   | 427   | 638   | (9)  |  |
|                                                                        | CDD10.0     | 1152        | 0     | 0     | 2     | 8     | 57    | 189   | 317   | 316   | 201   | 56    | 6     | 1     | (10) |  |
|                                                                        | CDD18.3     | 191         | 0     | 0     | 0     | 0     | 2     | 22    | 74    | 70    | 22    | 1     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 708         | 0     | 0     | 0     | 2     | 21    | 90    | 327   | 215   | 49    | 4     | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 150         | 0     | 0     | 0     | 0     | 4     | 14    | 81    | 42    | 9     | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 6.2         | 6.9   | 7.2   | 6.9   | 7.0   | 6.3   | 5.1   | 5.0   | 5.1   | 5.6   | 6.4   | 6.7   | 6.7   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 802         | 47    | 38    | 46    | 82    | 93    | 107   | 87    | 95    | 68    | 62    | 59    | 40    | (15) |  |
|                                                                        | PrecMax     | 1030        | 77    | 125   | 115   | 142   | 281   | 271   | 165   | 256   | 151   | 131   | 149   | 85    | (16) |  |
|                                                                        | PrecMin     | 658         | 9     | 7     | 8     | 28    | 26    | 24    | 36    | 36    | 6     | 20    | 11    | 7     | (17) |  |
|                                                                        | PrecStd     | 105         | 19    | 27    | 26    | 30    | 55    | 69    | 33    | 53    | 35    | 35    | 39    | 22    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 8.8   | 7.9   | 16.8  | 22.4  | 26.6  | 28.6  | 32.1  | 30.7  | 27.6  | 23.2  | 17.2  | 11.4  | (19) |  |
|                                                                        |             | MCWB        | 5.9   | 5.3   | 11.9  | 14.7  | 17.4  | 20.4  | 24.6  | 23.6  | 21.4  | 16.3  | 12.8  | 9.7   | (20) |  |
|                                                                        | 2%          | DB          | 5.7   | 5.2   | 11.4  | 15.8  | 21.4  | 25.9  | 28.9  | 27.6  | 24.5  | 19.2  | 14.0  | 8.1   | (21) |  |
|                                                                        |             | MCWB        | 4.0   | 3.7   | 7.3   | 10.8  | 15.6  | 20.6  | 22.0  | 22.1  | 20.2  | 16.1  | 11.4  | 6.6   | (22) |  |
|                                                                        | 5%          | DB          | 3.9   | 3.8   | 8.6   | 12.4  | 18.7  | 23.9  | 26.7  | 25.5  | 22.9  | 17.4  | 12.1  | 6.2   | (23) |  |
|                                                                        |             | MCWB        | 2.8   | 2.7   | 5.9   | 8.7   | 14.4  | 19.2  | 21.4  | 21.9  | 19.9  | 14.9  | 10.0  | 5.0   | (24) |  |
|                                                                        | 10%         | DB          | 2.5   | 2.6   | 6.3   | 10.0  | 16.2  | 21.9  | 24.9  | 24.1  | 21.7  | 15.8  | 10.4  | 4.8   | (25) |  |
|                                                                        |             | MCWB        | 1.7   | 1.7   | 4.4   | 7.1   | 12.9  | 18.5  | 21.0  | 21.6  | 19.2  | 13.7  | 8.7   | 3.8   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 6.6   | 6.2   | 13.3  | 15.2  | 20.1  | 23.7  | 26.3  | 26.0  | 23.5  | 19.0  | 13.7  | 10.3  | (27) |  |
|                                                                        |             | MCDB        | 7.2   | 7.6   | 14.8  | 19.8  | 22.2  | 25.5  | 27.4  | 28.0  | 24.5  | 20.6  | 16.1  | 11.2  | (28) |  |
|                                                                        | 2%          | WB          | 4.1   | 4.1   | 7.7   | 11.5  | 17.2  | 22.1  | 24.3  | 24.7  | 21.9  | 17.2  | 12.2  | 7.5   | (29) |  |
|                                                                        |             | MCDB        | 5.0   | 5.0   | 9.8   | 14.4  | 20.5  | 24.1  | 26.7  | 26.0  | 23.5  | 18.4  | 13.8  | 8.0   | (30) |  |
|                                                                        | 5%          | WB          | 2.8   | 2.9   | 5.6   | 9.1   | 15.0  | 20.6  | 23.0  | 23.6  | 20.9  | 15.6  | 10.6  | 5.3   | (31) |  |
|                                                                        |             | MCDB        | 3.6   | 3.7   | 7.0   | 11.1  | 17.5  | 22.2  | 25.1  | 24.6  | 22.5  | 16.7  | 11.8  | 5.9   | (32) |  |
|                                                                        | 10%         | WB          | 1.6   | 1.8   | 4.1   | 7.3   | 13.0  | 19.2  | 21.8  | 22.4  | 19.9  | 14.3  | 9.0   | 3.8   | (33) |  |
|                                                                        |             | MCDB        | 2.4   | 2.6   | 5.5   | 9.3   | 15.4  | 21.0  | 23.9  | 23.7  | 21.3  | 15.5  | 10.1  | 4.6   | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 6.2   | 5.9   | 5.6   | 5.6   | 6.4   | 6.4   | 6.1   | 5.2   | 5.7   | 5.6   | 5.5   | 5.7   | (35) |  |
|                                                                        |             | MCDBR       | 6.8   | 5.8   | 9.3   | 10.8  | 11.1  | 9.9   | 9.1   | 7.3   | 7.4   | 7.8   | 7.6   | 6.5   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 6.2   | 5.4   | 6.7   | 7.3   | 7.5   | 6.4   | 5.4   | 4.8   | 5.2   | 5.7   | 6.1   | 5.7   | (37) |  |
|                                                                        |             | MCWBR       | 5.8   | 5.7   | 8.0   | 10.1  | 10.4  | 8.7   | 8.0   | 6.1   | 6.4   | 6.6   | 6.8   | 5.8   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.286 | 0.314 | 0.338 | 0.385 | 0.409 | 0.427 | 0.419 | 0.413 | 0.388 | 0.345 | 0.309 | 0.286 | (40) |  |
|                                                                        |             | taud        | 2.447 | 2.349 | 2.385 | 2.297 | 2.286 | 2.286 | 2.312 | 2.322 | 2.376 | 2.488 | 2.522 | 2.490 | (41) |  |
|                                                                        | Ebn at Noon |             | 851   | 876   | 900   | 878   | 865   | 847   | 849   | 842   | 836   | 835   | 818   | 818   | (42) |  |
|                                                                        |             | Edh at Noon | 72    | 95    | 105   | 124   | 129   | 130   | 126   | 120   | 106   | 83    | 68    | 64    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1.46  | 2.40  | 3.59  | 4.54  | 5.38  | 5.99  | 6.14  | 5.25  | 4.20  | 2.65  | 1.64  | 1.15  | (44) |  |
|                                                                        | RadStd      |             | 0.19  | 0.27  | 0.34  | 0.43  | 0.45  | 0.42  | 0.27  | 0.34  | 0.34  | 0.29  | 0.15  | 0.12  | (45) |  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (47 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page